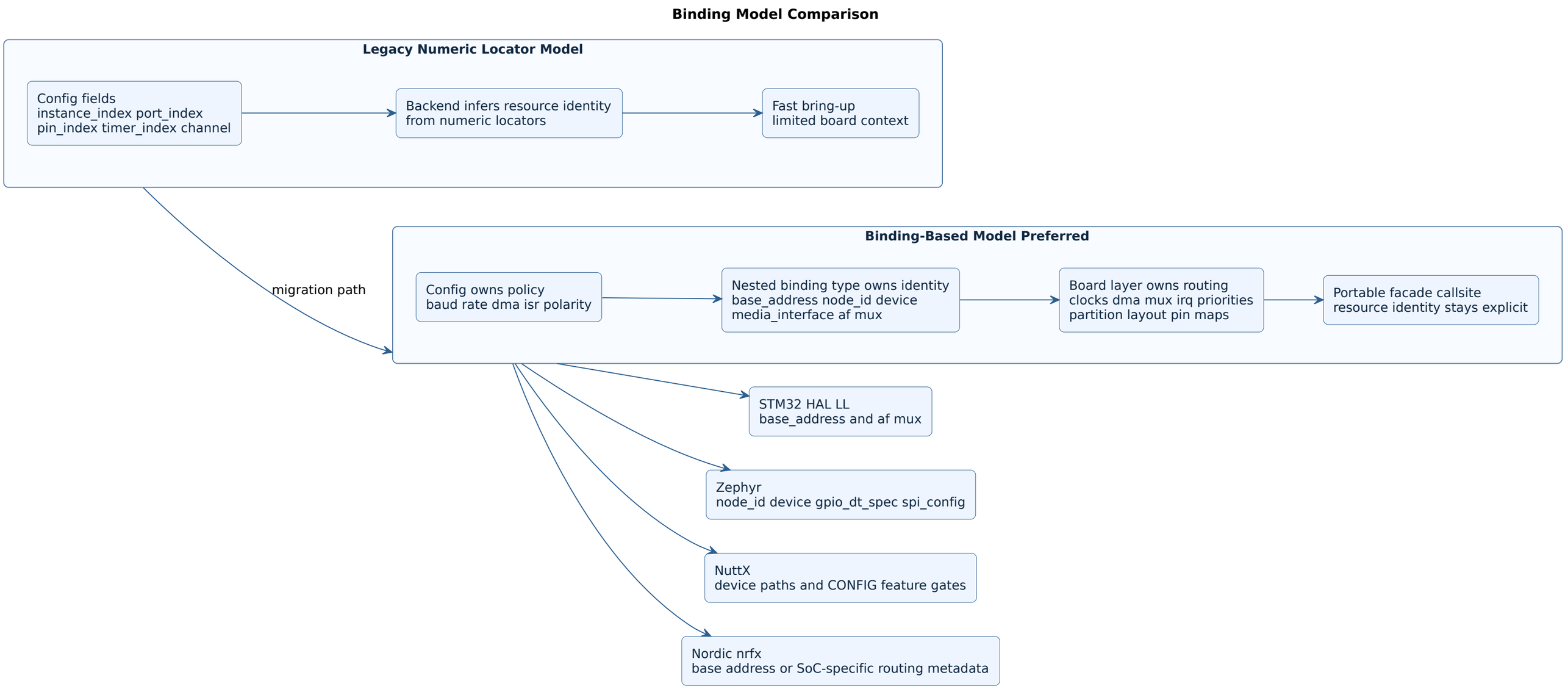
Binding Model Comparison
Legacy Numeric Locator Model
Config fields
instance_index port_index
pin_index timer_index channel
Backend infers resource identity
from numeric locators
Fast bring-up
limited board context
migration path
Binding-Based Model Preferred
Config owns policy
baud rate dma isr polarity
Nested binding type owns identity
base_address node_id device
media_interface af mux
Board layer owns routing
clocks dma mux irq priorities
partition layout pin maps
Portable facade callsite
resource identity stays explicit
STM32 HAL LL
base_address and af mux
Zephyr
node_id device gpio_dt_spec spi_config
NuttX
device paths and CONFIG feature gates
Nordic nrfx
base address or SoC-specific routing metadata